ExpNo-8 Conduct Z test for the data given using python

Program:

```
import numpy as np

import scipy.stats as stats

# Define the sample data (hypothetical weights in grams)

sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149,151, 150, 149, 152, 151,

148, 150, 152, 149, 150,148, 153, 151, 150, 149, 152, 148, 151, 150, 153])

# Population mean under the null hypothesis

population_mean = 152

# Calculate sample statistics

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1) # Using sample standard deviation

# Number of observations

n = len(sample_data)

# Calculate the Z-statistic

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))

# Calculate the p-value

p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic))) # Two-tailed test

# Print results

print(f"Sample Mean: {sample_mean:.2f}")

print(f"Z-Statistic: {z_statistic:.4f}")

print(f"P-Value: {p_value:.4f}")

# Decision based on the significance level

alpha = 0.05

if p_value < alpha:
```

```
    print("Reject the null hypothesis: The average weight is significantly different from 150 grams.")
else:
    print("Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.")
```

OUTPUT:

```
Sample Mean: 150.20
Z-Statistic: -5.7655
P-Value: 0.0000
Reject the null hypothesis: The average weight is significantly different from 150 grams.
```